

Manage Virtual Networking

VNets, subnets, NSGs, peering, DNS, service endpoints, private endpoints, and routing — the complete picture of how Azure resources communicate and how you control that traffic.

VNet structure — address space, subnets, resources

A Virtual Network is a private, isolated network in Azure. Everything inside it is invisible to the rest of the internet by default. You carve the VNet address space into subnets, then deploy resources into subnets — not directly into the VNet.

 **VNet — vnet-prod** 10.0.0.0/16 — 65,536 addresses

WEB TIER

10.0.1.0/24

VMs, App Service (VNet Integration), load balancer backend

APP TIER

10.0.2.0/24

App servers, container instances

DATA TIER

10.0.3.0/24

SQL, Storage private endpoints

GATEWAYSUBNET

10.0.255.0/27

VPN / ExpressRoute gateway — name must be exact

Azure reserves 5 addresses per subnet: .0 (network), .1 (default gateway), .2–.3 (DNS), .255 (broadcast). A /29 gives you only 3 usable IPs.

Eight terms you need cold

Virtual Network (VNet)

A logically isolated network in Azure. Resources in the same VNet can communicate by default. Traffic never leaves the Azure backbone unless you explicitly configure it to.

Tip: VNets are regional — they span all availability zones in a region but don't span regions. Use peering for cross-region connectivity.

Subnet

A range of IP addresses carved out of the VNet address space. Resources deploy into subnets, not directly into VNets. NSGs and route tables attach at the subnet level.

Tip: Some services require dedicated subnets (Azure Firewall, VPN Gateway, Bastion) — plan address space before deploying.

Network Security Group (NSG)

A stateful firewall consisting of inbound and outbound rules based on source/dest IP, port, and protocol. Attached to a subnet or NIC — both can apply to the same traffic.

Tip: NSG rules have priority 100–4096 — lower number wins. Default rules (65000–65500) allow VNet and Azure LB traffic; rule 65500 denies all remaining.

→ VNet Peering

Connects two VNets so their resources communicate using private IPs — no gateway, no public internet. Can be same-region (VNet peering) or cross-region (global peering).

Tip: Peering is non-transitive — if VNetA peers with VNetB and VNetB peers with VNetC, A and C cannot communicate without a direct peering or hub-spoke routing.

Azure DNS

Azure provides automatic DNS resolution within a VNet (using the private IP of the VM's hostname). Azure DNS Private Zones extend this with custom domain names across VNets.

Tip: Link a `sql.internal.contoso.com` to private DNS Zone to a VNet to resolve custom names like

Service Endpoint

Extends the VNet's identity to an Azure service (Storage, SQL, Key Vault) over the Azure backbone — the service firewall can then restrict access to that subnet only.

Tip: The service is still publicly addressed — only the traffic path changes. Use Private Endpoint for full private access.

Private Endpoint

A NIC with a private IP in your subnet that maps to an Azure PaaS service (Storage, SQL, Key Vault, etc). The service becomes fully private — unreachable from the internet when firewall is set to deny.

Tip: Requires a Private DNS Zone to resolve the service's FQDN to the private IP — without it, DNS still resolves to the public IP.

> User Defined Route (UDR)

A custom route table entry that overrides Azure's default system routes — used to force traffic through an NVA (Network Virtual Appliance) or Azure Firewall instead of going directly.

Tip: Attach the route table to the subnet — it overrides system routes for all traffic originating from that subnet.

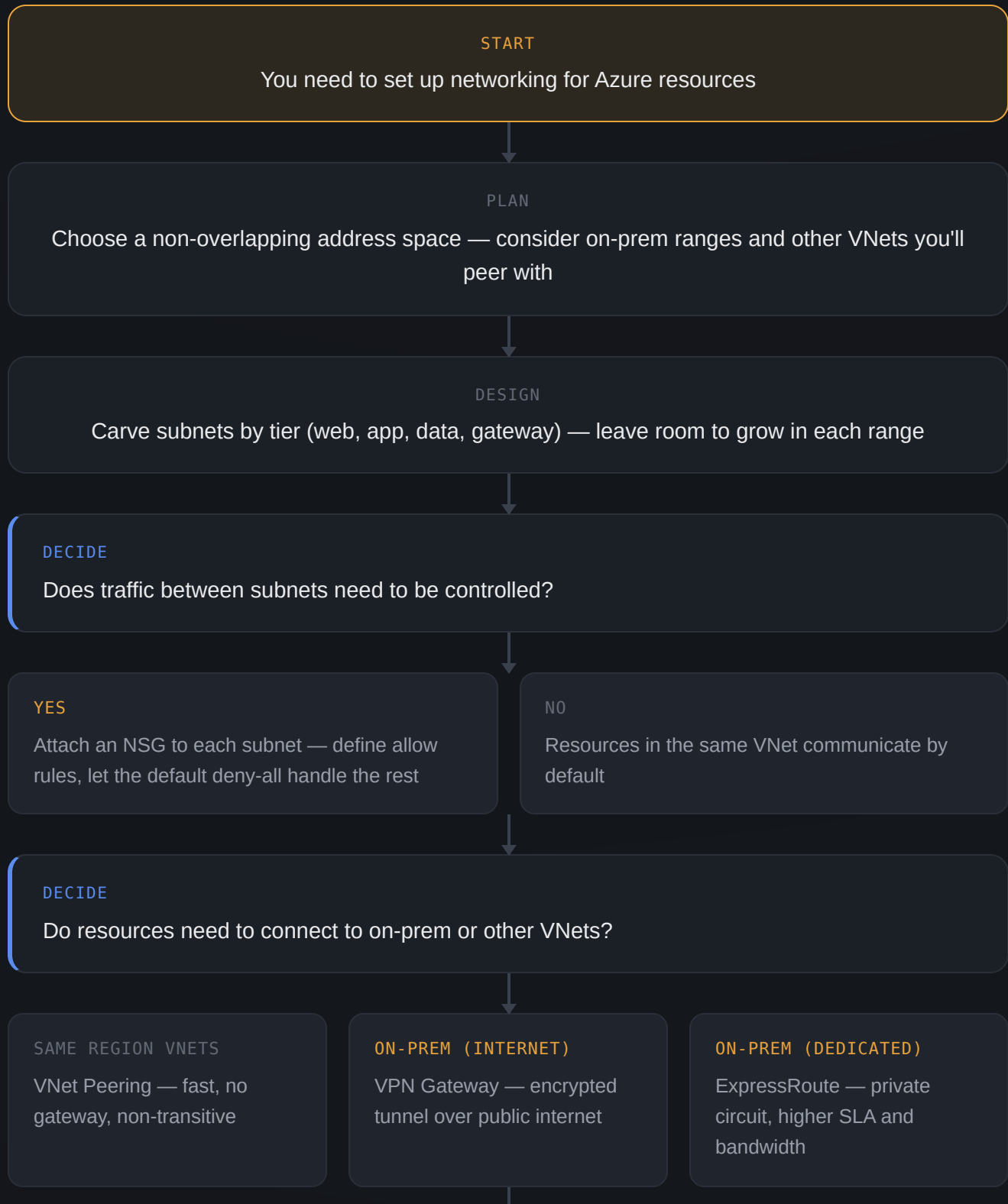
CONNECTIVITY OPTION	USE CASE	TRANSITIVE?	GATEWAY NEEDED?
VNet Peering	Same or cross-region VNet-to-VNet	No	No
VPN Gateway	On-prem to Azure over encrypted internet	Yes (hub-spoke)	Yes
ExpressRoute	On-prem to Azure over dedicated private circuit	Yes	Yes
Azure Virtual WAN	Large-scale hub-spoke across many sites	Yes	Yes

EXAM GOTCHA — PEERING IS NOT TRANSITIVE

If VNetA ↔ VNetB and VNetB ↔ VNetC, VNetA **cannot** reach VNetC without a direct peering between A and C — unless you route through a hub with **Allow gateway transit** enabled and use a VPN or ExpressRoute gateway in the hub. This is the hub-spoke pattern and it is tested heavily.

Designing and securing a virtual network

Walk through this when standing up a new VNet or connecting an existing one to on-premises or other VNets.



DECIDE

Do PaaS services (Storage, SQL, Key Vault) need private access?

FULL PRIVATE ACCESS

Private Endpoint + Private DNS Zone + deny public access on the service

RESTRICT SOURCE SUBNET ONLY

Service Endpoint — simpler but service stays publicly addressed

DECIDE

Does all outbound traffic need to go through a firewall or NVA?

YES

Create a UDR with next hop = Azure Firewall or NVA private IP, attach to affected subnets

NO

Default Azure routing handles it

DONE

VNet designed, subnets secured, services connected privately, routing enforced

Tools for virtual networking

>_ Azure CLI — VNet & Subnets

BEST FOR: scripted VNet creation and subnet management

Create VNets, add subnets, and configure DNS servers in commands that drop straight into a deployment script.

```
az network vnet create \
  -g rg-prod -n vnet-prod \
  --address-prefix 10.0.0.0/16 \
  --subnet-name web-subnet \
  --subnet-prefix 10.0.1.0/24

az network vnet subnet create \
  -g rg-prod --vnet-name vnet-prod \
  -n app-subnet \
  --address-prefix 10.0.2.0/24

az network vnet peering create \
  -g rg-prod -n peer-prod-dev \
  --vnet-name vnet-prod \
  --remote-vnet vnet-dev \
  --allow-vnet-access
```

>_ Azure CLI — NSG

BEST FOR: scripted security rule management

Create NSGs, add rules, and associate them with subnets in one script — essential for automating tiered network security.

```
az network nsg create \
  -g rg-prod -n nsg-web

az network nsg rule create \
  -g rg-prod --nsg-name nsg-web \
  -n allow-https --priority 100 \
  --protocol Tcp --direction
Inbound \
  --source-address-prefix '*' \
  --destination-port-range 443 \
  --access Allow

az network vnet subnet update \
  -g rg-prod --vnet-name vnet-prod \
  -n web-subnet \
  --network-security-group nsg-web
```

> Azure CLI — Private Endpoint & DNS

BEST FOR: making PaaS services fully private

Create a private endpoint for a storage account and wire up the DNS zone so the FQDN resolves to the private IP.

```
az network private-endpoint create \
  -g rg-prod -n pe-storage \
  --vnet-name vnet-prod \
  --subnet data-subnet \
  --private-connection-resource-id \
  $(az storage account show \
    -g rg-prod -n mystorageacct --
  query id -o tsv) \
  --group-id blob \
  --connection-name conn-storage

az network private-dns zone create \
  -g rg-prod \
  -n
  "privatelink.blob.core.windows.net"

az network private-dns link vnet
create \
  -g rg-prod \
  -n dns-link-prod \
  --zone-name
  "privatelink.blob.core.windows.net" \
  --virtual-network vnet-prod \
  --registration-enabled false
```

<> Bicep — VNet & NSG

BEST FOR: IaC — network topology as code

Deploy the entire network topology — VNet, subnets, NSGs, and associations — as a single version-controlled template.

```
resource vnet
'Microsoft.Network/virtualNetworks@
2023-04-01' = {
  name: 'vnet-prod'
  location: location
  properties: {
    addressSpace: {
      addressPrefixes: ['10.0.0.0/16'] }
    subnets: [
      {
        name: 'web-subnet'
        properties: {
          addressPrefix:
            '10.0.1.0/24'
          networkSecurityGroup: {
            id: nsg.id }
        }
      }
    ]
  }
}
```

> Azure CLI — Route Table (UDR)

BEST FOR: forcing traffic through a firewall or NVA

Create a route table with a default route pointing to Azure Firewall or an NVA, then associate it with subnets that need inspection.

```
az network route-table create \
  -g rg-prod -n rt-forced-tunnel

az network route-table route create \
  -g rg-prod --route-table-name rt-forced-tunnel \
  -n default-to-firewall \
  --address-prefix 0.0.0.0/0 \
  --next-hop-type VirtualAppliance \
  --next-hop-ip-address 10.0.0.4

az network vnet subnet update \
  -g rg-prod --vnet-name vnet-prod \
  -n app-subnet \
  --route-table rt-forced-tunnel
```

TOOL	BEST FOR	SUPPORTS PEERING?	REPEATABLE?
Portal	Visual topology, one-off changes	Yes	No
CLI	Scripting, automation	Yes	Yes
PowerShell	Windows enterprise automation	Yes	Yes
Bicep / ARM	IaC, full topology as code	Yes	Yes

Principles worth internalizing

FEATURE	SERVICE ENDPOINT	PRIVATE ENDPOINT
Service address	Still public IP	Private IP in your VNet
Traffic path	Azure backbone	Stays in VNet
Disables public access	No — firewall restricts source	Yes — when public access denied
DNS change required	No	Yes — Private DNS Zone needed
Cost	Free	Billed per hour + data processed

EXAM GOTCHA — NSG EVALUATION ORDER

When an NSG is attached to **both a subnet and a NIC**, inbound traffic is evaluated by the **subnet NSG first**, then the NIC NSG. Outbound traffic is evaluated by the **NIC NSG first**, then the subnet NSG. Both NSGs must allow the traffic — one deny is enough to block it.

PRIVATE ENDPOINT DNS — THE MOST COMMON SETUP MISTAKE

When you create a Private Endpoint for a storage account, DNS still resolves `mystorageacct.blob.core.windows.net` to its **public IP** until you create a **Private DNS Zone** (`privatelink.blob.core.windows.net`) and link it to the VNet. Without the DNS zone, traffic bypasses the private endpoint entirely.

Six mistakes that show up on the exam

Assuming peering is transitive

A ↔ B and B ↔ C does not mean A ↔ C. You need a direct peering between A and C, or a hub with gateway transit enabled.

Forgetting Azure reserves 5 IPs per subnet

A /29 gives 8 addresses, minus 5 reserved = only 3 usable. Plan subnets larger than you think you need — you can't resize a subnet that contains resources.

Using overlapping address spaces for peered VNets

VNet peering fails if the address spaces overlap — even partially. Plan all VNet ranges before deployment.

Setting up a Private Endpoint without the DNS zone

Without a Private DNS Zone linked to the VNet, DNS resolves the service to its public IP and traffic bypasses the private endpoint entirely.

Confusing Service Endpoint with Private Endpoint

Service Endpoints keep the service publicly addressed — just restrict the source. Private Endpoints give the service a private IP and can make it fully unreachable from the internet.

Forgetting NSG evaluation order (subnet then NIC inbound)

Inbound: subnet NSG first, then NIC NSG.
Outbound: NIC NSG first, then subnet NSG. A deny at either layer blocks the traffic.